

Appendix

1. Introduction

This document serves as an appendix to the paper titled “Evaluating active learning methods for error correction”¹. This appendix includes the results from the model-feature hyperparameter grid search in Section 2, the classification results in Section 3, the regression results in Section 4, and a summary of the results in Section 5.

2. Grid search results

Hyperparameters are selected following a ten-fold cross-validation grid search for each model-feature combination. Grid searches are conducted on all available data, withholding data used for the holdout sets. The results for the decision tree, k-NN, and MLP algorithms are shown in Tables A1, A2, and A3, respectively. For all tables, a problem type of R represents regression and C represents classification.

Table A1: Decision Tree Grid Search Results

Datas et	Feature	Proble m Type (R/C)	Criterion	Max Dep th	Min Sampl es Leaf	Splitt er	Metric	Bes t Sco re
HVB	Material Weight	R	squared_er ror	10	1	best	R ²	1.00
	Material Part Number	R	squared_er ror	10	1	best	R ²	0.99
	Minimu m Capacit y	R	squared_er ror	15	1	rando m	R ²	1.00
	Capacit y Through put	R	absolute_e rror	20	1	best	R ²	1.00
	Voltage	R	friedman_ mse	7	32	best	R ²	0.75
	Model Code	C	gini	10	1	rando m	Accura cy	1.00

¹ Shen, D.N., Basson, A.H., Grobler, J.: Evaluating active learning methods for error correction (2026). Submitted to the South African Computer Journal.

Datas et	Feature	Proble m Type (R/C)	Criterion	Max Dep th	Min Sampl es Leaf	Splitt er	Metric	Bes t Sco re
Red wine- qualit y	Quality	C	gini	15	1	best	Accura cy	0.62
Abalo ne	Rings	R	squared_er ror	10	32	best	R ²	0.60

For the k-NN algorithm (Table A2), the cosine distance metric is selected for most features and shows significantly better training performance than other distance metrics in the conducted experiments.

Table A2: k-NN Grid Search Results

Datase t	Feature	Proble m Type (R/C)	Metric (Distanc e)	Neighbou rs	Weight s	Metric	Best Scor e
HVB	Material Weight	R	cosine	3	distan ce	R ²	1.00
	Material Part Number	R	cosine	3	distan ce	R ²	0.92
	Minimum Capacity	R	cosine	3	distan ce	R ²	0.99
	Capacity Throughp ut	R	cosine	3	distan ce	R ²	0.76
	Voltage	R	cosine	11	unifor m	R ²	0.71
	Model Code	C	manhatt an	3	distan ce	Accura cy	0.70
Red wine- quality	Quality	C	cosine	11	distan ce	Accura cy	0.66
Abalon e	Rings	R	cosine	15	distan ce	R ²	0.59

The results for the MLP algorithm are shown in Table A3. The adaptive moment estimation (Adam) solver was selected for all MLP-based active learners, as the datasets are large [31]. The maximum iterations were tuned to 2000 to balance model fit and computational effort. L2 regularisation was tested separately, and an ‘alpha’ value of 0.0001 was selected.

Table A3: MLP Grid Search Results

Dataset	Feature	Problem Type (R/C)	Activation	Hidden Layers	Learning Rate	Metric	Best Score
HVB	Material Weight	R	logistic	(130, 50)	0.00	R ²	0.91
	Material Part Number	R	relu	(70, 10)	0.00	R ²	0.37
	Minimum Capacity	R	logistic	(100, 70)	0.00	R ²	0.94
	Capacity Throughput	R	identity	(150, 10)	0.01	R ²	0.21
	Voltage	R	identity	(90, 50)	0.01	R ²	0.32
	Model Code	C	logistic	(70, 10)	0.00	Accuracy	0.82
Red wine-quality	Quality	C	logistic	(50, 130)	0.00	Accuracy	0.57
Abalone	Rings	R	logistic	(80, 30)	0.00	R ²	0.27

3. Additional classification results

Figure A1 shows the sample performance of the three model types on the “Quality” feature from the red wine dataset.

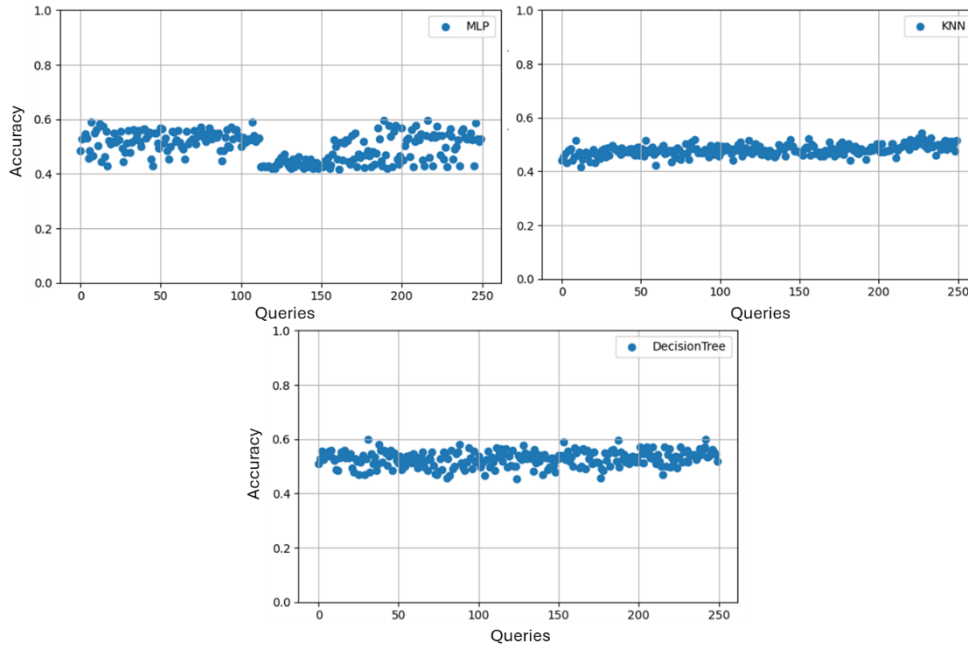


Figure A1: Sample distributions of model performance for the accuracy on the “Quality” feature, grouped by model type

In Figure A1, the MLP algorithm exhibits a large decrease in accuracy after 100 queries (top-left), the KNN algorithm shows robust performance and minor improvement in accuracy (<2%) as the active learning process continues (top-right), and the decision tree algorithm exhibits robust performance similar to a random distribution (bottom).

The decision tree learners generally outperform other model types, exhibiting robust performance. While model types such as the k-NN improve as more queries are retrieved, the performance does not reach that of the decision tree models that are trained on the initial training data. Although models such as k-NN improved as more queries were added, their performance did not reach that of decision trees trained on the initial dataset. Model performance can be improved by removing noise from the training sets, such as noise created by querying similar data.

3.1 Red wine – quality

Table A4 shows the mean accuracy results for the active learners on the red wine dataset.

Table A4: Accuracy results on the red wine quality dataset over 30 repetitions

Learner (Query Strategy)	Accuracy (%)	Standard Deviation (%)
Decision tree (QBC)	53.42	0.31
Decision tree (Least confidence)	53.39	0.25
Decision tree (Margin sampling)	53.33	0.28

Learner (Query Strategy)	Accuracy (%)	Standard Deviation (%)
Decision tree (Random)	53.32	0.30
Decision tree (Entropy sampling)	53.13	0.52
MLP (random)	49.38	0.37
MLP (Least Confidence)	49.21	0.41
MLP (Margin Sampling)	49.19	0.42
MLP (Entropy Sampling)	49.15	0.38
MLP (QBC)	48.88	1.29
KNN (QBC)	48.56	0.72
KNN (Entropy Sampling)	48.51	0.69
KNN (Least Confidence)	48.50	0.89
KNN (random)	48.49	0.99
KNN (Margin Sampling)	48.37	0.79

The Decision tree (QBC and least confidence) learners are tied for first place based on the statistical significance. The entropy sampling yields the lowest performance across model types on this feature. Table A5 shows the Mann-Whitney U test results.

Table A5: Summarised Mann-Whitney U test results for model-strategy pairs

Learner (Query Strategy)	Total Wins	Total Losses	Total Draws	Performance Score
Decision tree (QBC)	11	0	3	0.89
Decision tree (Least confidence)	11	0	3	0.89
Decision tree (Margin sampling)	10	0	4	0.86
Decision tree (Random)	10	0	4	0.86
Decision tree (Entropy sampling)	10	2	2	0.79
MLP (Random)	6	5	3	0.54
MLP (QBC)	5	5	4	0.50
MLP (Least confidence)	5	5	4	0.50
MLP (Margin sampling)	5	5	4	0.50
MLP (Entropy sampling)	5	6	3	0.46
KNN (QBC)	0	10	4	0.14
KNN (Least confidence)	0	10	4	0.14
KNN (Random)	0	10	4	0.14

KNN (Margin sampling)	0	10	4	0.14
KNN (Entropy sampling)	0	10	4	0.14

Figure A2 shows the accuracy, as a percentage, across all active learners and per model type for the “Quality” feature. The QBC strategy consistently has a better performance, with a better distribution and higher maximum values. The entropy sampling strategy shows the opposite, with relatively longer tails and less robust performance.

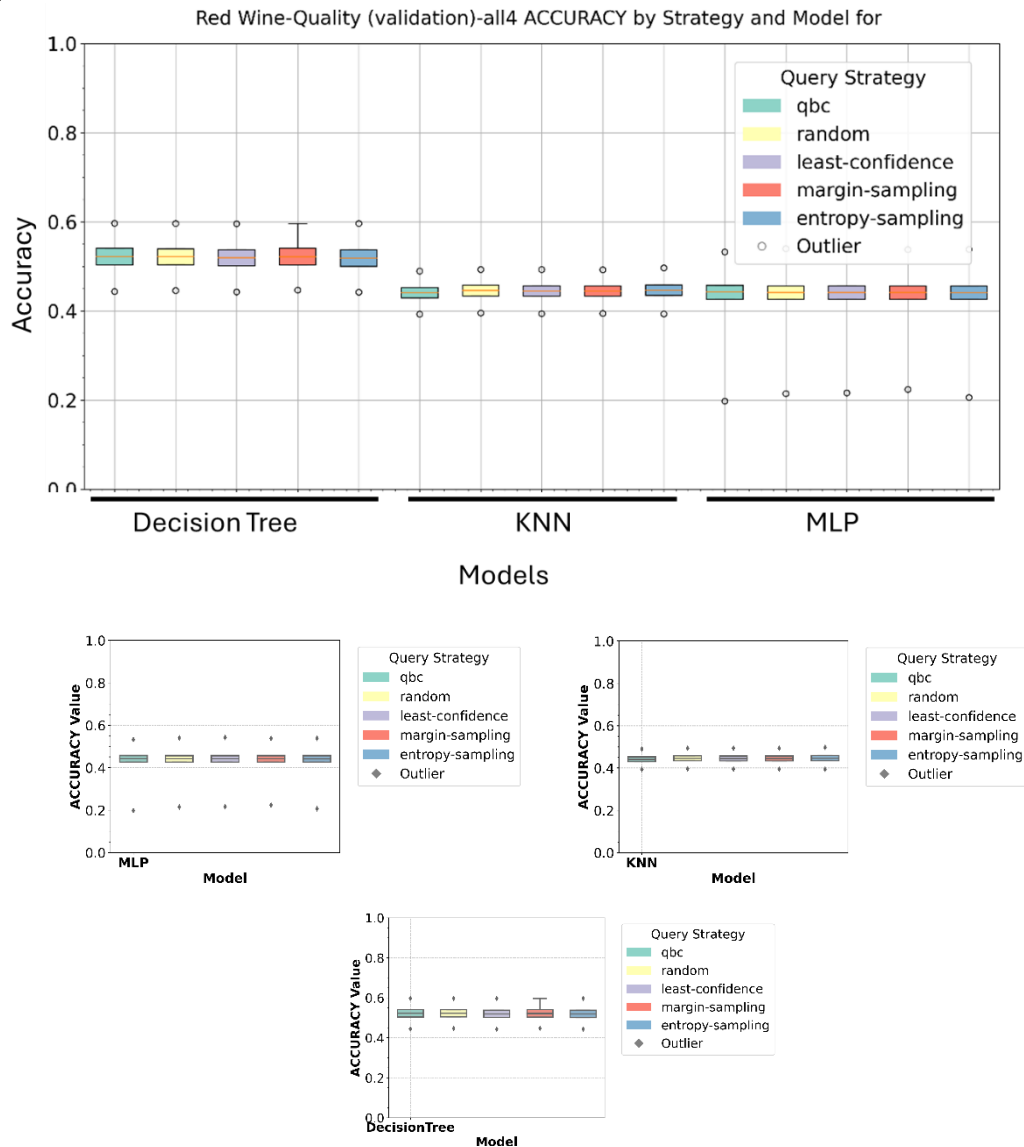


Figure A2: Accuracy results from all active learners (top) and per model type (bottom) on the Quality feature from the Red wine dataset

The decision tree learners generally outperform other model types, exhibiting robust performance. While model types such as the k-NN improve as more queries are retrieved, the performance does not reach that of the decision tree

models that are trained on the initial training data. Although models such as k-NN improved as more queries were added, their performance did not reach that of decision trees trained on the initial dataset. Model performance can be improved by removing noise from the training sets, such as noise created by querying similar data.

3.2 Model Code

Table A6 shows the accuracy and standard deviation for each active learner.

Table A6: Mean accuracy and standard deviation over 30 repetitions on the 'Model Code' feature of the HVB dataset

Learner (Strategy)	Accuracy (%)	Standard Deviation (%)
Decision Tree (Random)	98.94	0.07
Decision Tree (Least Confidence)	98.84	0.29
Decision Tree (QBC)	98.83	0.34
Decision Tree (Margin Sampling)	98.82	0.35
Decision Tree (Entropy Sampling)	98.82	0.27
KNN (Entropy Sampling)	62.88	0.28
KNN (QBC)	62.87	0.25
KNN (Least Confidence)	62.87	0.30
KNN (Margin Sampling)	62.84	0.25
KNN (Random)	62.83	0.19
MLP (QBC)	58.43	1.82
MLP (Entropy Sampling)	58.38	1.60
MLP (Margin Sampling)	57.91	1.45
MLP (Random)	57.69	1.75
MLP (Least Confidence)	57.59	1.63

The decision tree (Random) learner dominates in this dataset, being the only statistically significant result of its model type, with the QBC learners of the k-NN and MLP have slightly better performance in their specific model types, but

results are not found to be statistically significant. Table A7 shows the Mann-Whitney U test results for the “Model Code” feature.

Table A7: Mann-Whitney U-test results for the accuracy metric on the Model Code feature from the HVB dataset

Learner (Strategy)	Total Wins	Total Losses	Total Draws	Performance Score
Decision tree (Random)	11	0	3	0.89
Decision tree (QBC)	10	0	4	0.86
Decision tree (Margin sampling)	10	0	4	0.86
Decision tree (Least confidence)	10	0	4	0.86
Decision tree (Entropy sampling)	10	1	3	0.82
KNN (QBC)	5	5	4	0.50
KNN (Margin sampling)	5	5	4	0.50
KNN (Least confidence)	5	5	4	0.50
KNN (Random)	5	5	4	0.50
KNN (Entropy sampling)	5	5	4	0.50
MLP (QBC)	0	10	4	0.14
MLP (Margin sampling)	0	10	4	0.14
MLP (Least confidence)	0	10	4	0.14
MLP (Random)	0	10	4	0.14
MLP (Entropy sampling)	0	10	4	0.14

Figure A3 shows the accuracy, as a percentage, across all active learners and per model type for the “Model Code” feature. The random sampling seems to show more robust performance in this dataset. The QBC performance seems to be less robust in this feature, indicating a relatively even distribution of class instances, attributed to the larger dataset size, as compared to the red wine dataset.

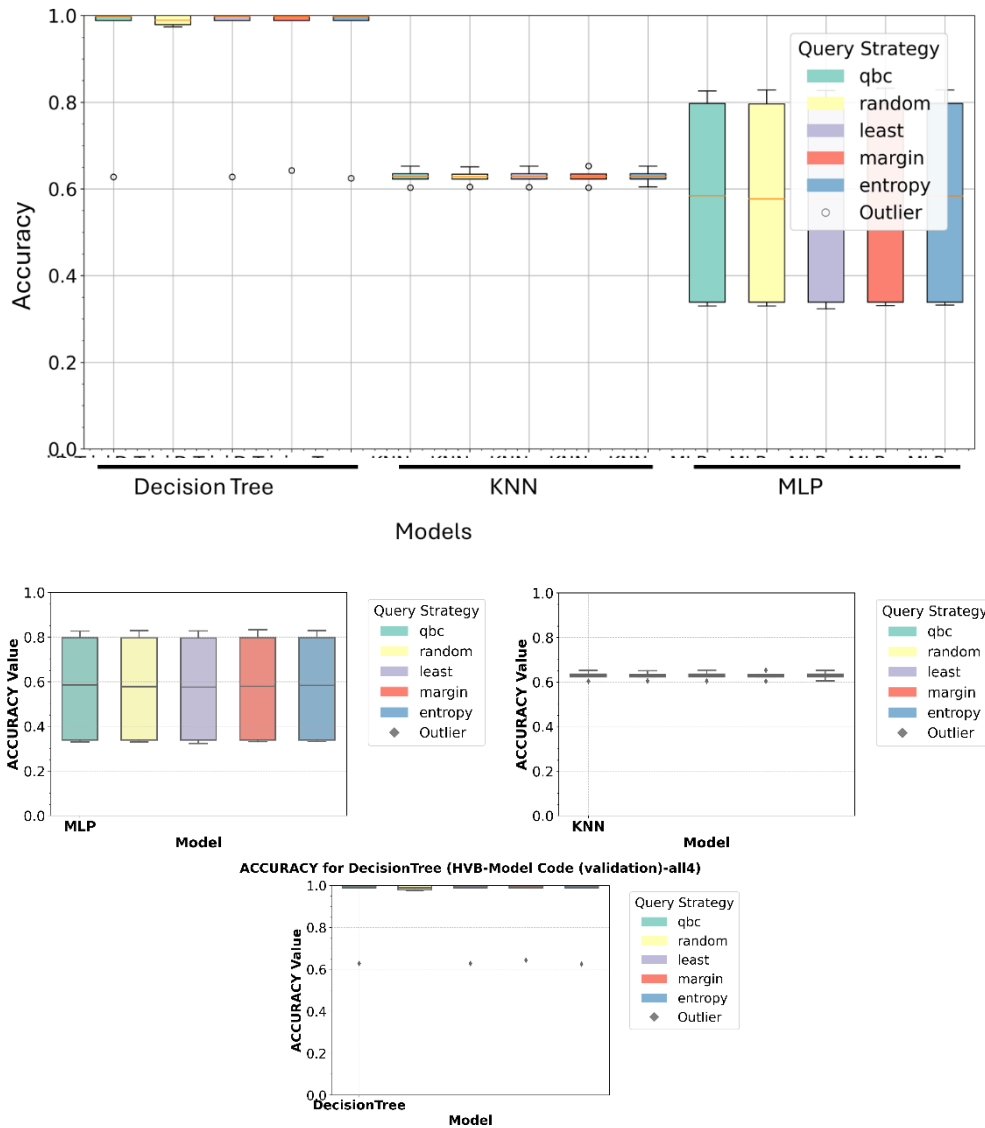


Figure A3: Accuracy results from all active learners (top) and per model type (bottom) on the Model Code feature from the HVB dataset

4. Additional regression results

NRMSE, NMAE, and R^2 distributions for each model type are included under the relevant feature subheading.

On the HVB dataset, QBC outperformed random sampling across most models and features. Decision trees achieved the highest R^2 scores (≥ 0.99) and the lowest NRMSEs on “Minimum Capacity” and “Capacity Throughput” (Tables 5 and 6), with QBC again slightly outperforming random sampling. KNN also performed well, while MLPs had the lowest model performance. The rankings are consistent across all metrics for each feature.

4.1 Abalone – Rings

Table A8 shows the results for the “rings” feature on the abalone dataset.

Table A8: Mean results on the Rings feature from the abalone dataset over 30 repetitions

Learner (Strategy)	NRMSE	Standard Deviation	R²	Standard Deviation
Decision Tree (QBC)	0.08	0.00	0.53	0.01
Decision Tree (random)	0.08	0.00	0.52	0.01
KNN (QBC)	0.08	0.00	0.52	0.00
KNN (random)	0.08	0.00	0.51	0.00
MLP (QBC)	0.15	0.00	-0.91	0.16
MLP (random)	0.15	0.01	-0.96	0.24

Decision tree (QBC) achieved the highest performance score, winning all of its pairwise comparisons, while MLPs underperformed across both strategies. Notable is the KNN (QBC) learner tied in second place with the decision tree (Random) learner. Table 9 shows the Mann-Whitney U results.

Table A9: Mann-Whitney U-test results rankings on the Rings feature from the abalone dataset

Model (Strategy)	Total Wins	Total Losses	Total Draws	Performance Score
Decision tree (QBC)	5	0	0	1.00
KNN (QBC)	3	1	1	0.70
Decision tree (random)	3	1	1	0.70
KNN (Random)	2	3	0	0.40
MLP (QBC)	0	4	1	0.10
MLP (Random)	0	4	1	0.10

Figures of NRMSE, NMAE, and R² are shown in Figures A4, A5, and A6, respectively.

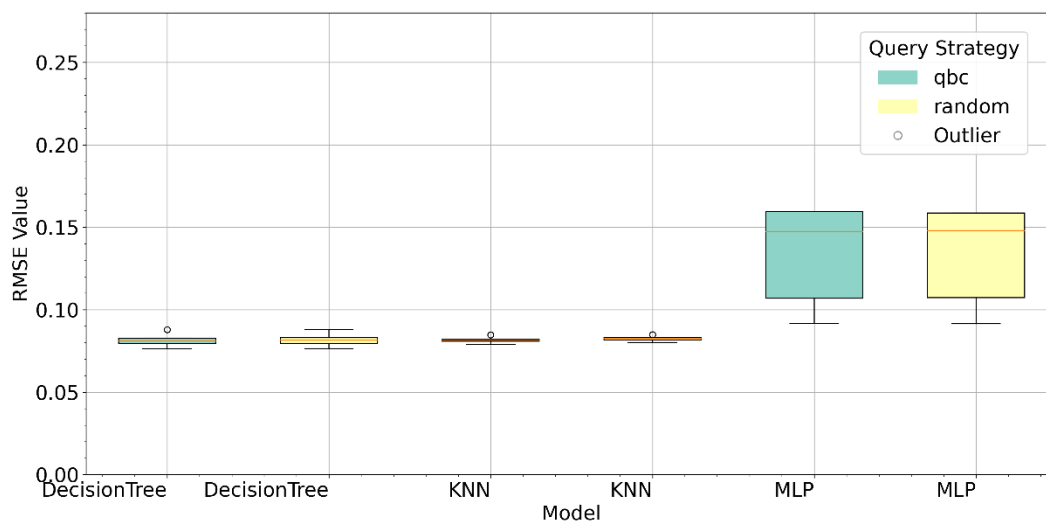


Figure A4: NRMSE results for all active learners on the “Rings” feature.

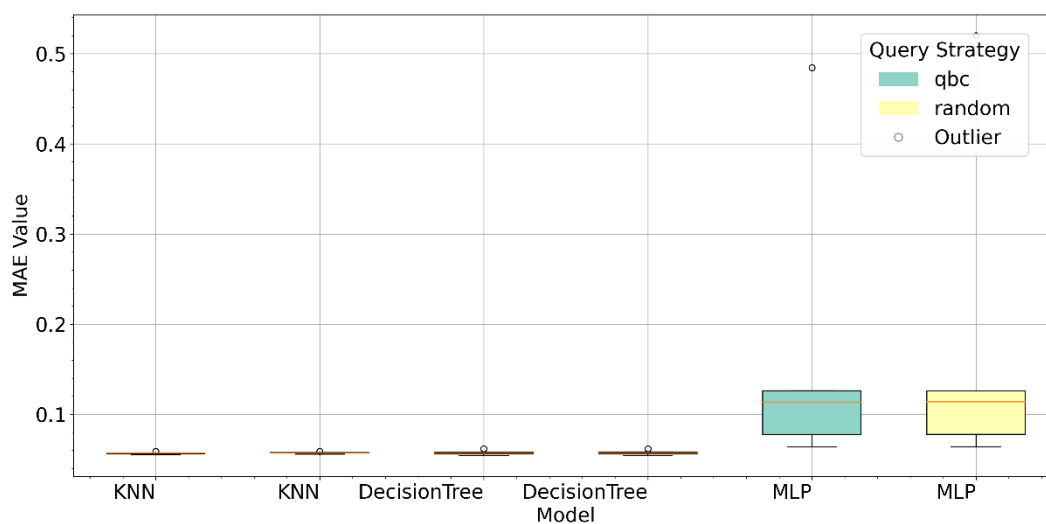


Figure A5: NMAE results for all active learners on the “Rings” feature.

Figure A6 shows the R^2 scores for the active learners on the abalone dataset.

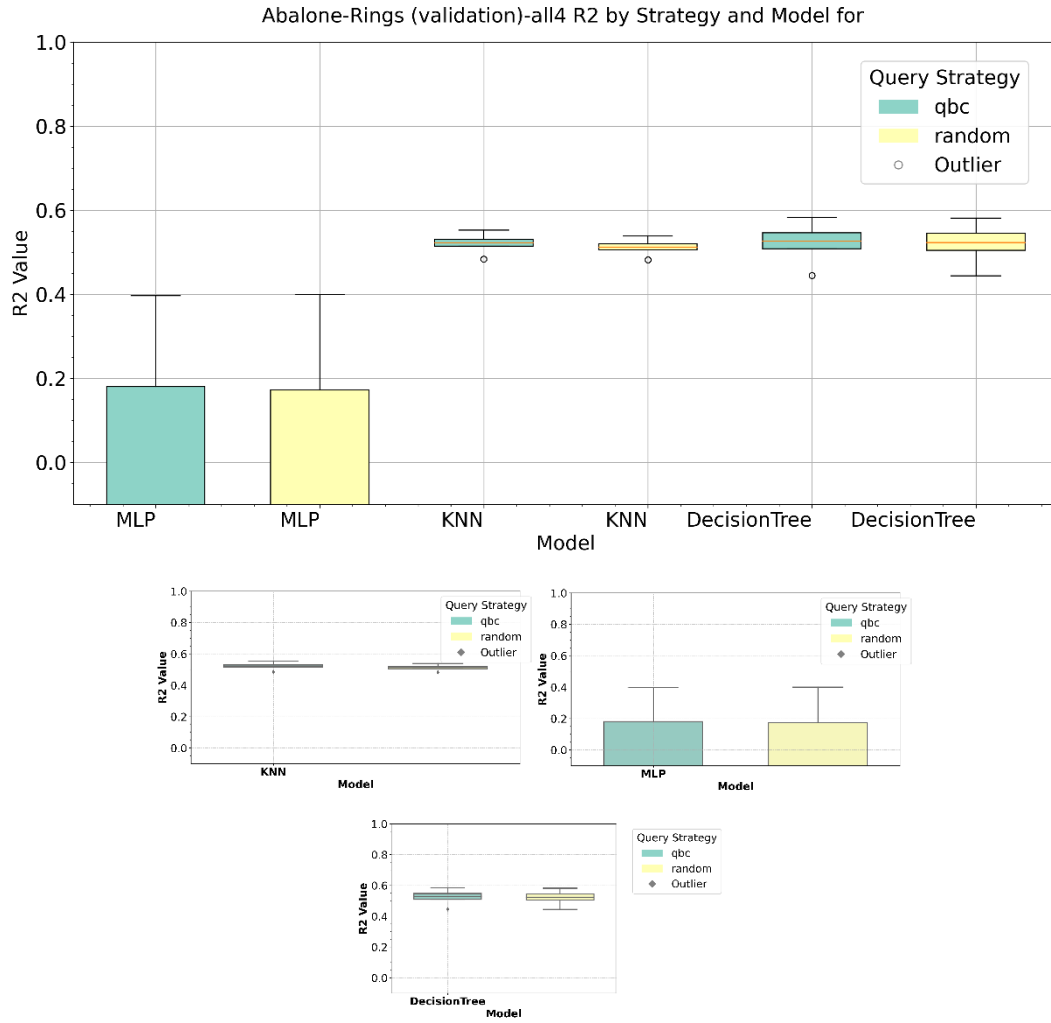


Figure A6: R^2 results from all active learners (top) and per model type (bottom) on the Rings feature from the abalone dataset

4.2 Capacity Throughput

Table A10 shows the results of the active learners on the “Capacity Throughput” feature of the HVB dataset

Table A10: Mean results on Capacity Throughput

Learner (Strategy)	NRMSE	Standard deviation	R^2	Standard deviation
Decision Tree (QBC)	0.01	0.00	0.99	0.00
Decision Tree (random)	0.01	0.00	0.99	0.00
KNN (QBC)	0.05	0.00	0.71	0.00
KNN (Random)	0.05	0.00	0.70	0.00
MLP (QBC)	0.10	0.00	-0.31	0.05

Learner (Strategy)	NRMSE	Standard deviation	R ²	Standard deviation
MLP (Random)	0.10	0.00	-0.35	0.08

Decision tree (QBC) achieved the top performance score, outperforming all other learners in its pairwise comparisons. The random strategy version of the decision tree also performed strongly, ranking second. MLPs, particularly under random sampling, consistently underperformed. KNN (QBC) achieved moderate performance but was outperformed by both decision tree variants. The Mann-Whitney U results are shown in Table A11.

Table A11: Mann-Whitney U-test results rankings on the Capacity Throughput feature from the HVB dataset

Model (Strategy)	Total Wins	Total Losses	Total Draws	Performance Score
Decision tree (QBC)	5	0	0	1.00
Decision tree (Random)	4	1	0	0.80
KNN (QBC)	3	2	0	0.60
KNN (Random)	2	3	0	0.40
MLP (QBC)	1	4	0	0.20
MLP (Random)	0	5	0	0.00

Figures of NRMSE, NMAE, and R² are shown in Figure 7, Figure A8, and Figure A9, respectively. There is a clear difference between the QBC variants and the random sampling variants, with statistical significance confirmed for each model type.

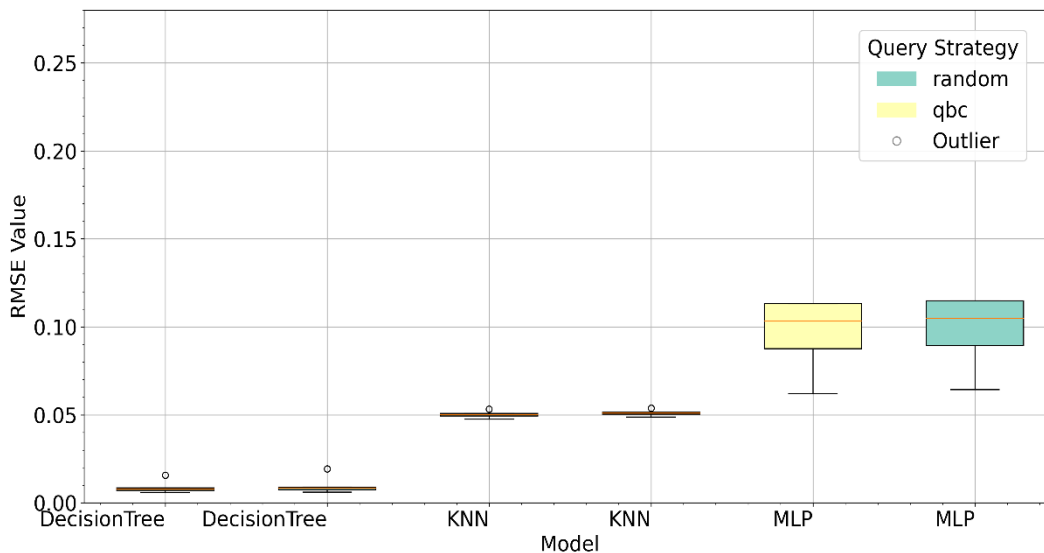


Figure 7: NRMSE results for all active learners on the “Capacity Throughput” feature.

The NMAE and R^2 results for all active learners are shown in Figure A8 and Figure A9, respectively.

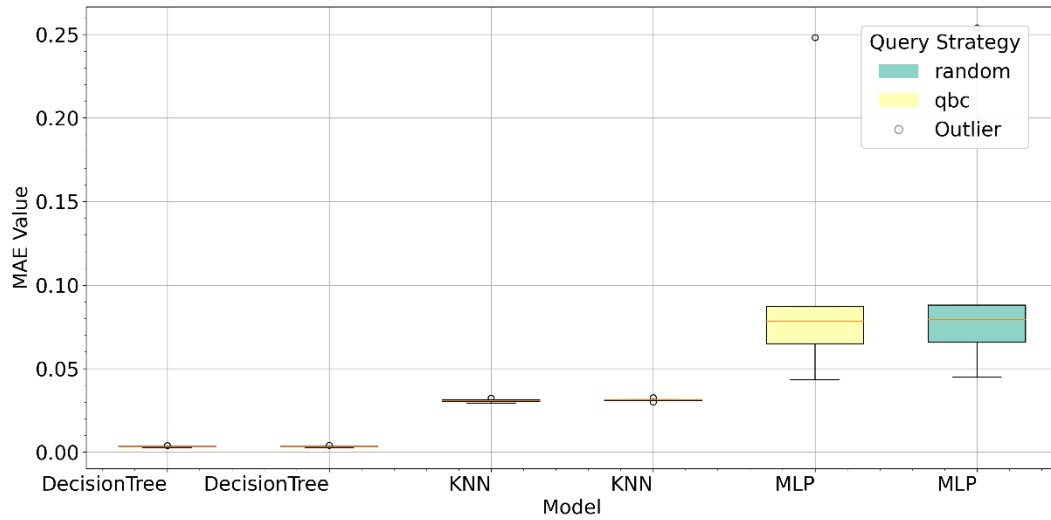


Figure A8: NMAE results for all active learners on the “Capacity Throughput” feature

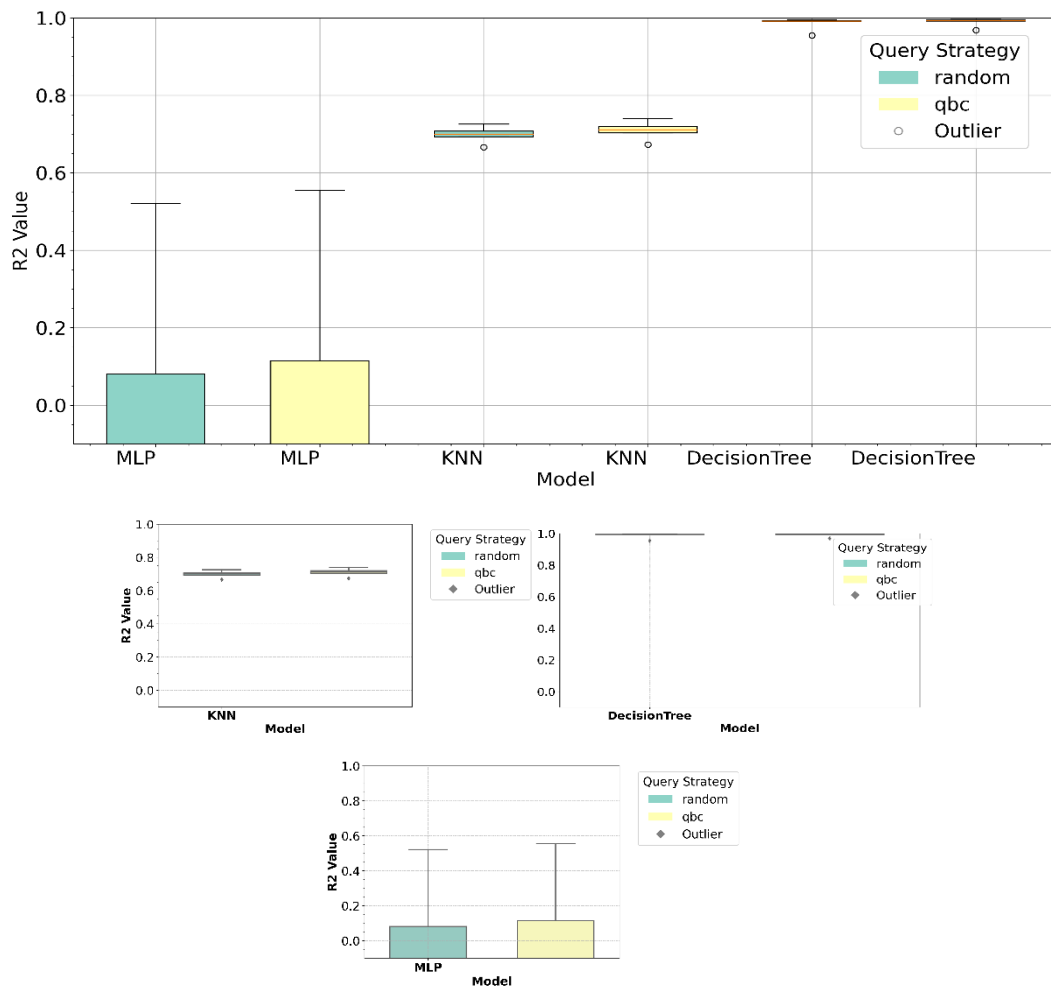


Figure A9: R^2 results from all active learners (top) and per model type (bottom) on the Capacity Throughput feature from the HVB dataset.

4.3 Minimum Capacity

Table A12 contains the results for the active learners on the “*Minimum Capacity*” feature of the HVB dataset.

Table A12: Mean results on minimum capacity

Learner (Strategy)	NRMSE	Standard deviation	R ²	Standard Deviation
Decision Tree (QBC)	0.00	0.00	1.00	0.00
Decision Tree (random)	0.00	0.00	1.00	0.00
KNN (QBC)	0.02	0.00	0.99	0.00
KNN (Random)	0.02	0.00	0.99	0.00
MLP (Random)	0.11	0.01	0.76	0.03
MLP (QBC)	0.11	0.01	0.75	0.04

Decision tree (Random) and decision tree (QBC) learners tied for the highest performance score, each winning four pairwise comparisons and drawing once. KNN (QBC) outperformed its random counterpart, while both MLP variants underperformed, with identical scores. These results demonstrate consistency in decision tree robustness across querying strategies. Mann-Whitney U tests are shown in Table A13.

Table A13: Mann-Whitney U-test ranking results on the Minimum Capacity feature from the HVB dataset

Learner (Strategy)	Total Wins	Total Losses	Total Draws	Performance Score
Decision tree (Random)	4.00	0.00	1.00	0.90
Decision tree (QBC)	4.00	0.00	1.00	0.90
KNN (QBC)	3.00	2.00	0.00	0.60
KNN (Random)	2.00	3.00	0.00	0.40
MLP (Random)	0.00	4.00	1.00	0.10
MLP (QBC)	0.00	4.00	1.00	0.10

The distributions of NMAE, NRMSE, and R^2 are shown in Figure A10, Figure A11, and Figure A12

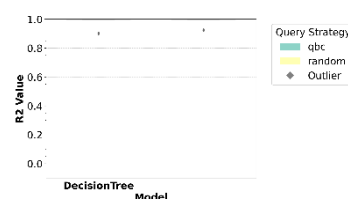


Figure A12, respectively. Only the querying k-NN (QBC) learner outperforms its random sampling variant. The MLP (Random) learn yields lower error and a better R^2 score, but this is not confirmed to be statistically significant.

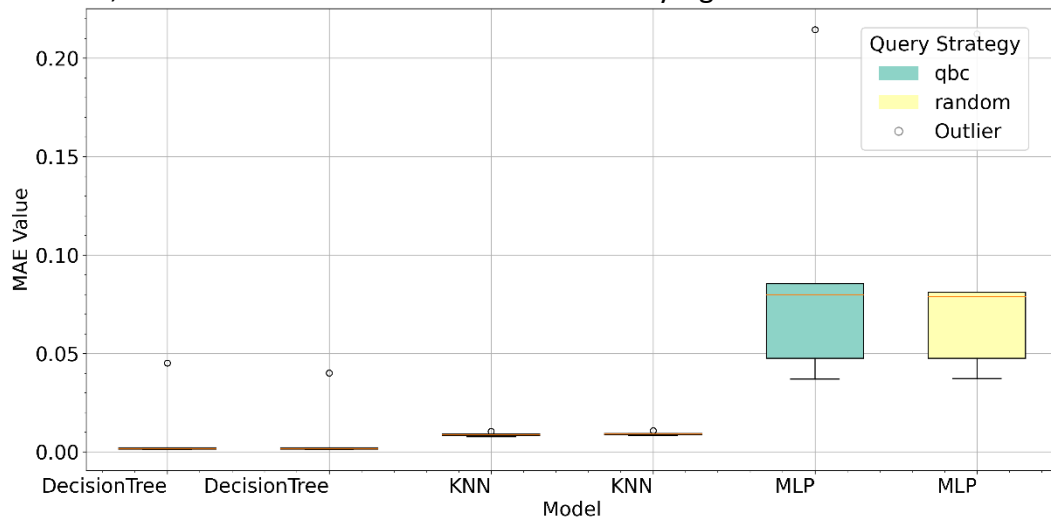


Figure A10: NMAE results for all active learners on the “Minimum Capacity” feature.

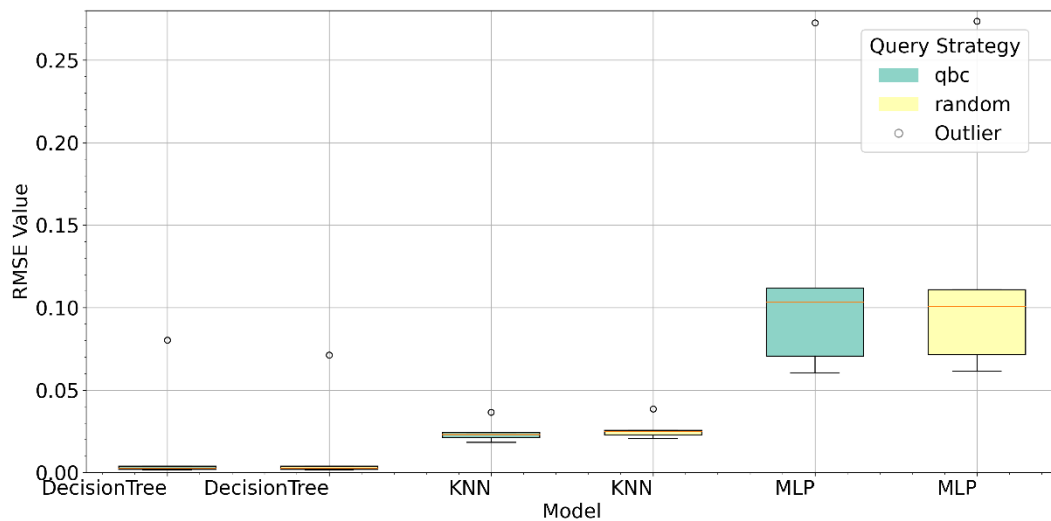


Figure A11: NRMSE results for all active learners on the “Minimum Capacity” feature.



Figure A12 presents the R^2 results. On the Minimum Capacity feature.

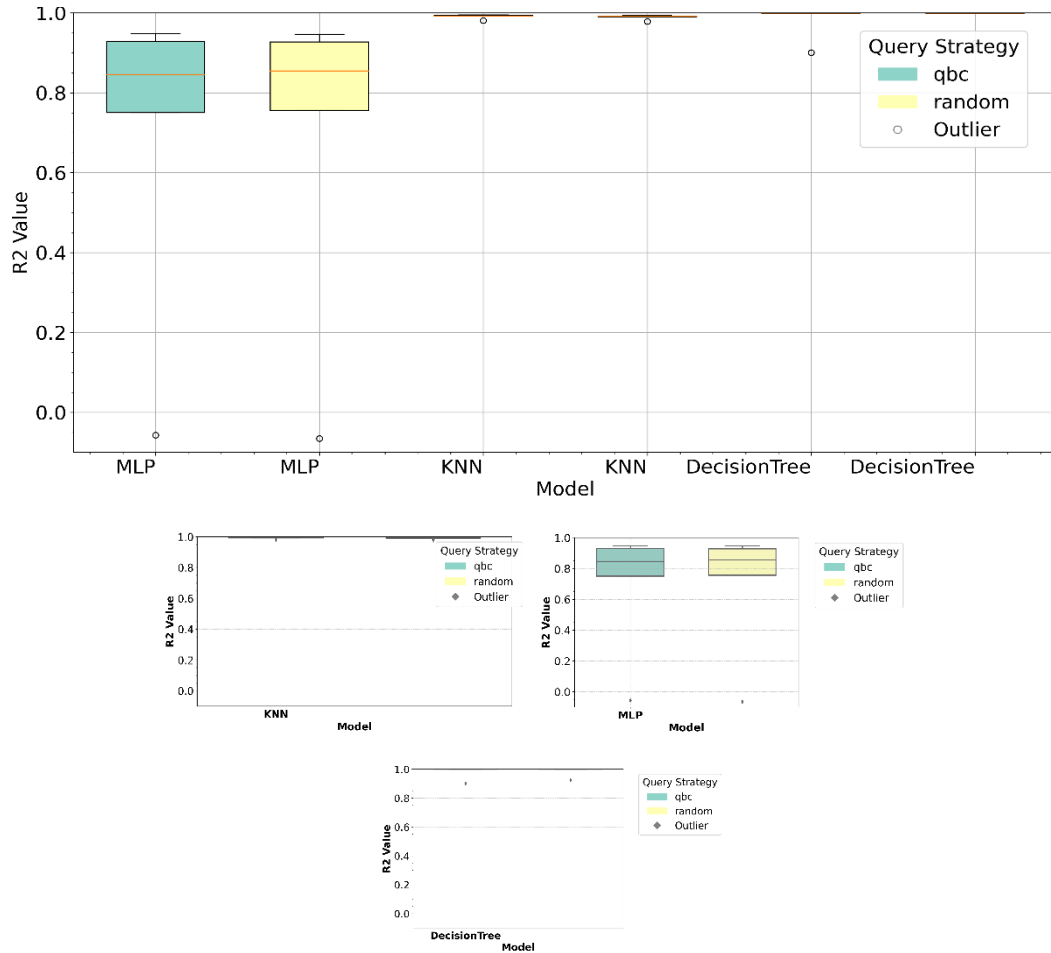


Figure A12: R^2 results from all active learners (top) and per model type (bottom) on the Minimum Capacity feature from the HVB dataset.

4.4 Voltage

The results of the active learners on the “Voltage” feature from the HVB dataset are shown in Table A14.

Table A14: Mean results on voltage

Learner (Query Strategy)	NRMSE	Standard deviation	R^2	Standard deviation
Decision Tree (Random)	0.10	0.00	0.71	0.00
Decision Tree (QBC)	0.10	0.00	0.71	0.00

KNN (QBC)	0.11	0.00	0.68	0.00
KNN (Random)	0.11	0.00	0.67	0.00
MLP (QBC)	0.18	0.00	0.06	0.02
MLP (Random)	0.18	0.00	0.05	0.02

Decision tree learners, using both QBC and random sampling, tied for the top performance score, each winning four comparisons and drawing one. KNN (QBC) again outperformed KNN (Random), consistent with previous features. The MLP (QBC) learner outperforms its random counterpart. These findings further highlight the reliability of decision trees and the added value of QBC for k-NN and MLP. Mann-Whitney U results are shown in Table A15.

Table A15: Mann-Whitney U-test results rankings on the Voltage feature from the HVB dataset

Learner (Strategy)	Total Wins	Total Losses	Total Draws	Performance Score
Decision tree (QBC)	4	0	1	0.90
Decision tree (Random)	4	0	1	0.90
KNN (QBC)	3	2	0	0.60
KNN (Random)	2	3	0	0.40
MLP (QBC)	1	4	0	0.20
MLP (Random)	0	5	0	0.00

Figure A13, Figure A14, and Figure A15 show the NRMSE, NMAE, and R^2 distributions, respectively.

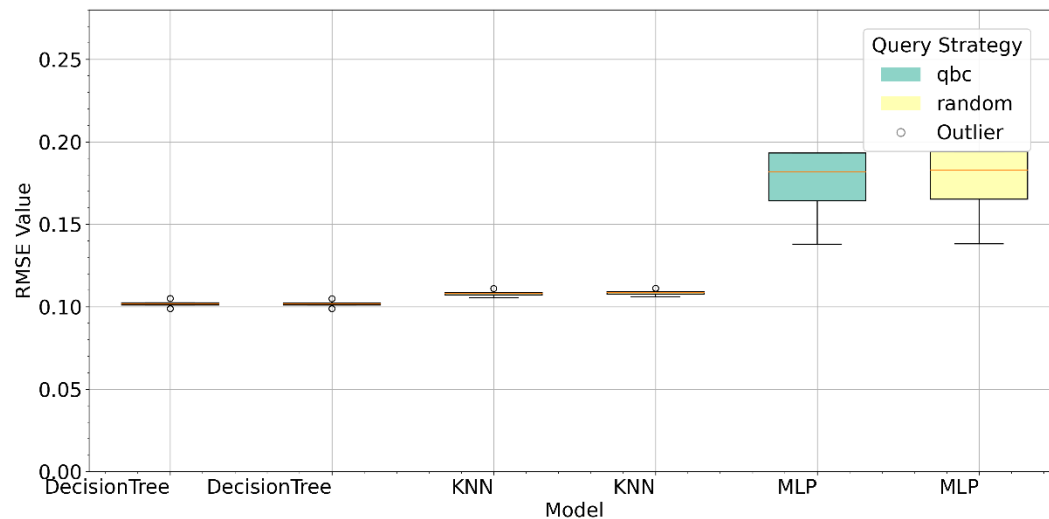


Figure A13: NRMSE results for all active learners on the “Voltage” feature.

Figure A14 shows NMAE, while Figure A15 shows R² score

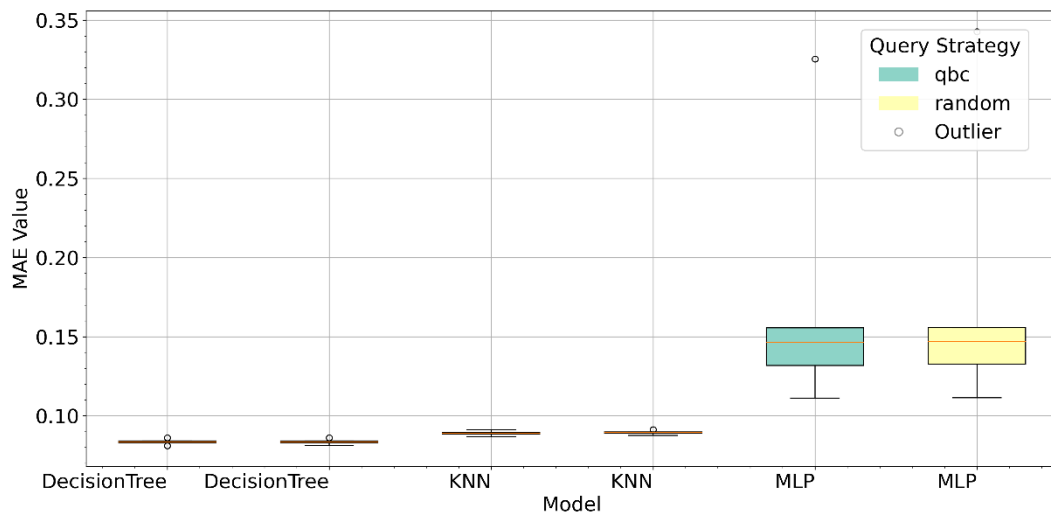


Figure A14: NMAE for the “Voltage” feature

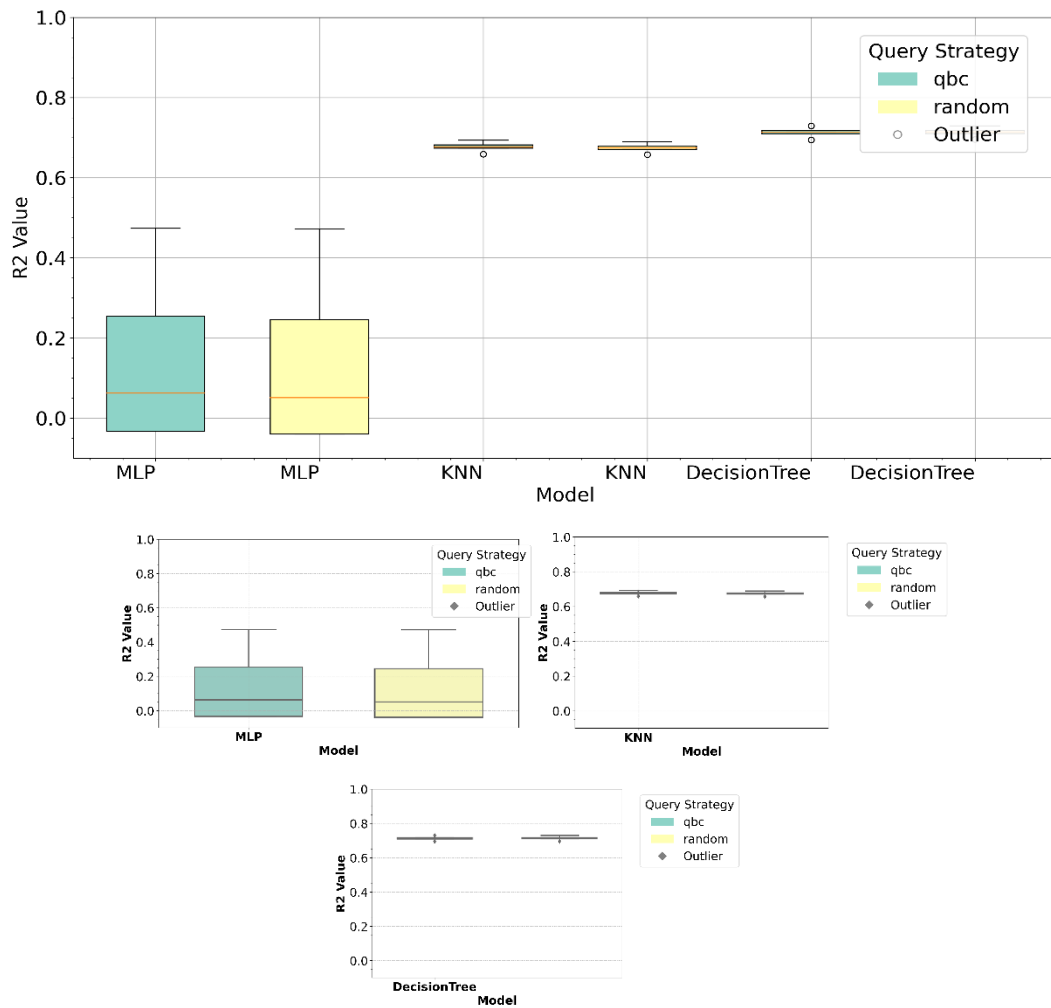


Figure A15: R² results from all active learners (top) and per model type (bottom) on the Voltage feature from the HVB dataset

Figure A15 shows the clear improvement in R^2 score due to the QBC strategy.

4.5 Material Weight

The results on the Material Weight feature are shown in Table A16.

Table A16: Mean results on Material Weight

Learner (Strategy)	NRMSE	Standard deviation	R^2	Standard deviation
KNN (QBC)	0.02	0.00	0.99	0.00
KNN (Random)	0.02	0.00	0.99	0.00
Decision Tree (Random)	0.11	0.00	0.85	0.00
Decision Tree (QBC)	0.11	0.00	0.85	0.00
MLP (QBC)	0.15	0.01	0.62	0.05
MLP (Random)	0.15	0.01	0.59	0.05

The Mann-Whitney U-test results are shown in Table A17.

Table A17: Mann-Whitney U-test ranking results on the “Material Weight” feature from the HVB dataset

Learner (Strategy)	Total Wins	Total Losses	Total Draws	Performance Score
KNN (QBC)	5	0	0	1.00
KNN (Random)	4	1	0	0.80
Decision Tree (QBC)	2	2	1	0.50
Decision Tree (Random)	2	2	1	0.50
MLP (QBC)	1	4	0	0.20
MLP (Random)	0	5	0	0.00

Figure A16, Figure A17, and Figure A18 present the NRMSE, NMAE, and R^2 distributions, respectively. The improvements caused by the QBC strategy are evident in model types, except for the decision tree learners.

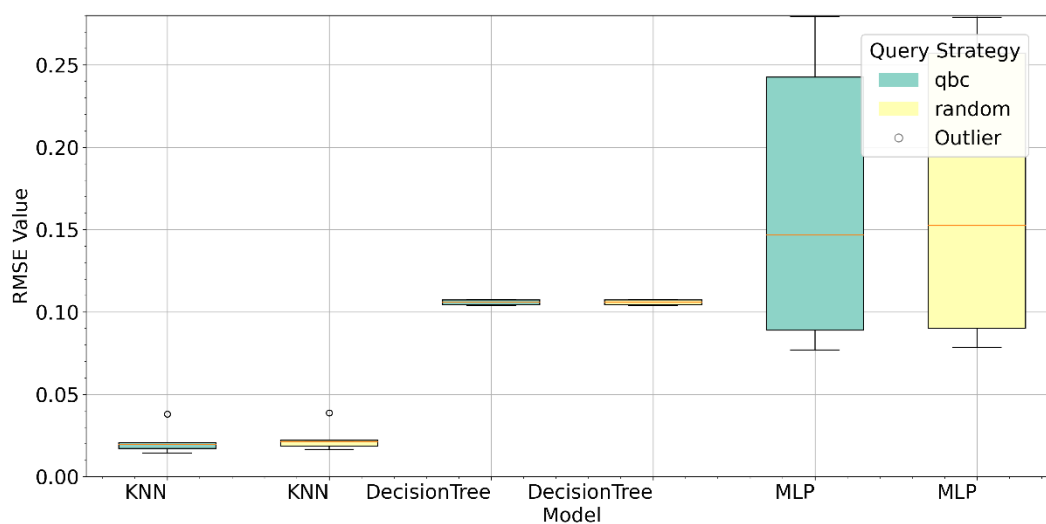


Figure A16: NRMSE results for all active learners on the “Material Weight” feature.

Figure A17 and Figure A18 present the NMAE, and R^2 distributions, respectively.

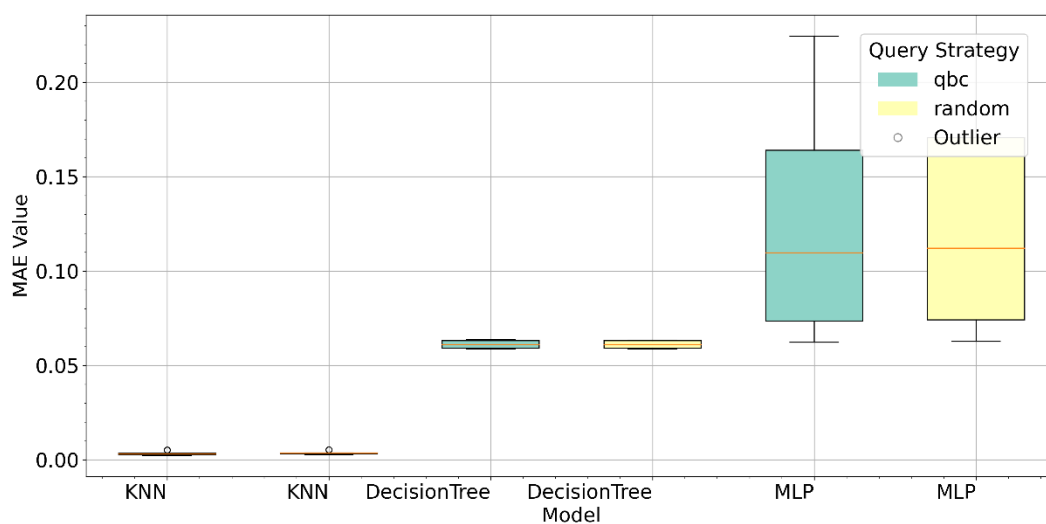


Figure A17: NMAE results for all active learners on the “Material Weight” feature.

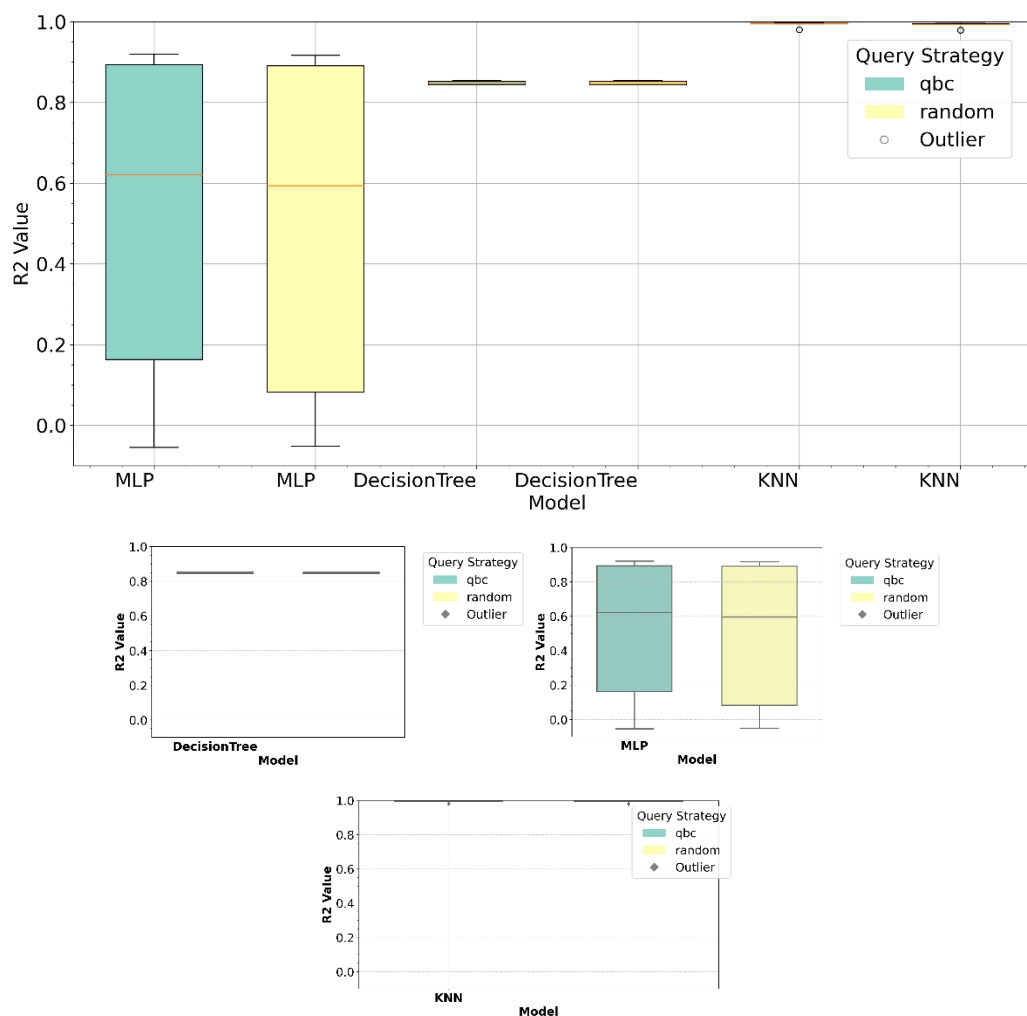


Figure A18: R² results from all active learners (top) and per model type (bottom) on the Material Weight feature from the HVB dataset

4.6 Material Part Number

Table A18 presents the results of the active learners on the Material Part Number feature.

Table A18: Mean results on Material Part Number

Learner (Strategy)	NRMSE	Standard deviation	R ²	Standard deviation
Decision Tree (QBC)	0.06	0.00	0.92	0.06
Decision Tree (random)	0.06	0.00	0.92	0.06
KNN (QBC)	0.09	0.00	0.81	0.09
KNN (Random)	0.09	0.00	0.81	0.09
MLP (QBC)	0.27	0.01	-1.66	0.27
MLP (Random)	0.28	0.02	-2.70	0.28

Similar to the “*Capacity Throughput*” feature, the decision tree and k-NN QBC learners outperform their random sampling variants. The MLP learners exhibit low performance on this feature, seen by the negative R^2 scores. Results of the Mann-Whitney U test are shown in Table A19.

Table A19: Mann-Whitney U-test ranking results on the “Material Part Number” feature from the HVB dataset

Model (Strategy)	Total Wins	Total Losses	Total Draws	Performance Score
Decision tree (QBC)	5	0	0	1.00
Decision tree (Random)	4	1	0	0.80
KNN (QBC)	3	2	0	0.60
KNN (Random)	2	3	0	0.40
MLP (QBC)	0	4	1	0.10
MLP (Random)	0	4	1	0.10

The NMAE, NRMSE, and R^2 scores are shown in Figure A19, Figure A20, and Figure A21, respectively.

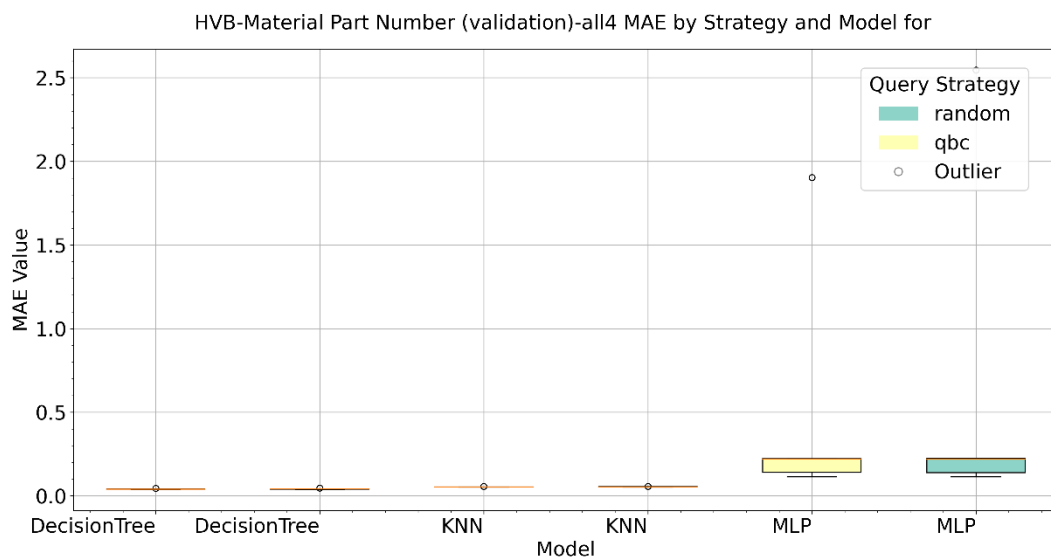


Figure A19: NMAE results for all active learners on the “Material Part Number” feature.

The NRMSE, and R^2 scores are shown in Figure A20, and Figure A21, respectively.

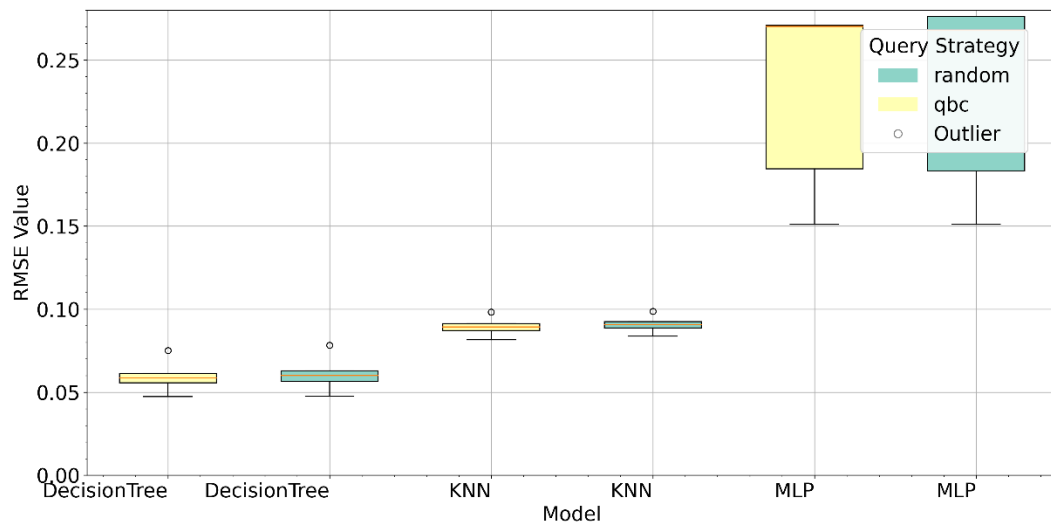


Figure A20: NRMSE results for all active learners on the “Material Part Number” feature.

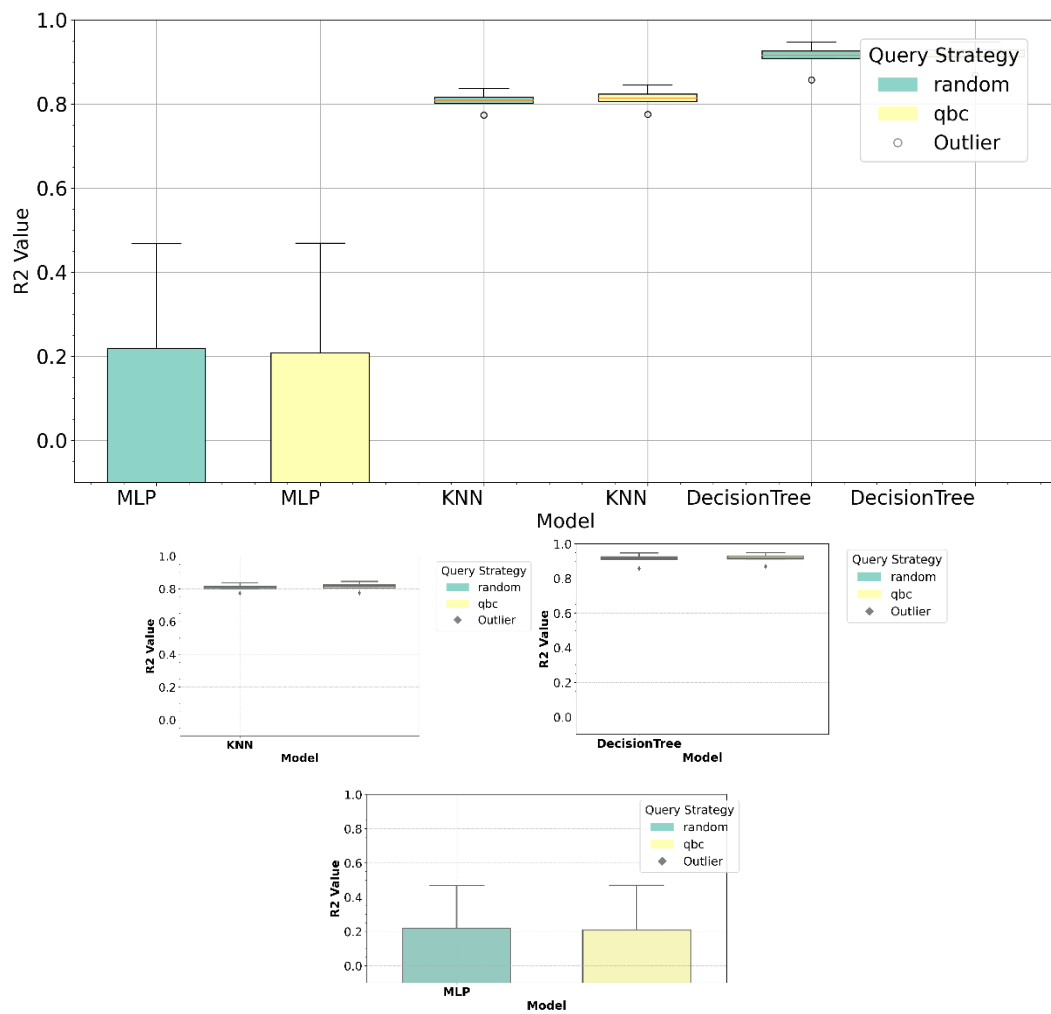


Figure A21: R^2 results from all active learners (top) and per model type (bottom) on the Material Part Number feature from the HVB dataset

5. Summary of Results

The extended experiments confirm that the effectiveness of an active learning strategy depends on both the task and data characteristics. Across datasets and features, decision tree-based active learners consistently achieved the highest accuracy and lowest error, demonstrating strong generalisation and robustness to incremental updates. The query-by-committee (QBC) strategy generally improved model performance—particularly for weaker learners such as MLPs—though its advantage diminished in balanced or low-variance datasets where random sampling performed comparably.

Results also show that k-NN learners benefited most from QBC in relative terms, especially on features with smoother or more continuous relationships, such as Material Weight. However, the marginal gains were modest compared to those of the decision tree learners. MLP learners exhibited high sensitivity to newly added queries, with the QBC strategy mitigating but not eliminating the observed sensitivity.

On the HVB dataset, QBC outperformed random sampling across most models and features. Decision trees achieved the highest R^2 scores (≥ 0.99) and the lowest NRMSEs on “Minimum Capacity” and “Capacity Throughput” (Table A12 and Table A14), with QBC again slightly outperforming random sampling. KNN also performed well, while MLPs showed limited effectiveness despite optimisation. The rankings are consistent across all metrics for each feature. Each features learner rankings, NRMSE, NMAE, and R^2 distributions and distributions for R^2 for each model type are included under the relevant subheading.

On the “Material Part Number” feature (Table A20), decision trees again outperformed other learners, with QBC producing slightly better results than random sampling. MLP models exhibited performance, with strongly negative R^2 values, suggesting poor fit.

Across all regression experiments, the decision tree (QBC) active learner consistently demonstrated the best overall performance, outperforming other learners on three of the six features and matching the decision tree (random) learner on two. The k-NN (QBC) learner achieved the highest performance on “Material Weight” and was joint second-best on the “Rings” feature of the abalone dataset.

Overall, the appendix findings reinforce the conclusions of the main study: decision tree and k-NN active learners offer a practical balance between performance, interpretability, and retraining efficiency, while QBC provides a reliable but computationally more complex strategy for improving generalisation.